

GURTEEN, Ireland

WMO: 039660

Lat: 53.017N

Lon: 8.000W

Elev: 249

StdP: 14.56

Time Zone: 1.00 (IRL)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.6	27.0	21.7	16.4	26.3	24.3	18.6	29.2	30.4	46.8	27.9	46.8	3.1	40	0.636

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.8	73.6	62.9	70.2	61.7	67.5	60.2	64.6	71.1	63.0	68.3	61.6	66.3	8.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.1	84.3	67.0	60.9	80.6	65.5	59.4	76.4	63.9	29.7	71.4	28.6	68.2	27.6	66.4	70.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.3	22.3	19.6	20.9	78.0	4.8	4.1	17.4	81.0	14.6	83.4	11.8	85.7	8.3	88.7		
			20.4	67.4	4.9	2.1	16.9	68.8	14.0	70.1	11.3	71.2	7.8	72.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.2	40.8	41.4	42.8	46.9	51.6	56.5	59.7	58.1	55.2	50.3	44.1	41.8
	DBStd	8.42	5.75	5.50	5.14	4.82	4.90	4.30	3.93	3.31	4.08	4.95	5.69	7.00
	HDD50	1425	286	245	228	114	37	3	0	0	5	57	189	261
	HDD65	5793	750	662	687	542	415	257	170	214	293	456	627	720
	CDD50	1118	2	3	6	22	87	198	302	252	162	66	12	6
	CDD65	10	0	0	0	0	0	2	7	1	0	0	0	0
	CDH74	107	0	0	0	0	3	42	61	1	0	0	0	0
	CDH80	14	0	0	0	0	0	10	4	0	0	0	0	0
Wind	WSAvg	9.5	10.6	10.4	9.9	9.7	9.3	8.3	8.2	8.9	9.1	9.2	9.7	10.9
Precipitation	PrecAvg	37.1	3.8	2.8	2.6	2.6	2.5	2.8	3.1	3.1	2.8	4.0	3.8	3.7
	PrecMax	44.4	6.4	7.2	4.4	4.5	5.6	5.2	5.6	6.8	6.1	6.3	8.9	7.0
	PrecMin	29.0	1.2	0.6	0.7	0.6	0.3	0.7	1.7	0.4	1.0	1.1	1.9	1.5
	PrecStd	4.1	1.4	1.8	1.0	1.1	1.3	1.3	0.9	1.7	1.3	1.7	1.6	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.3	55.5	60.6	65.5	73.8	80.6	80.3	72.0	70.8	64.3	58.9	56.7
		MCWB	53.3	51.4	50.9	56.7	62.7	62.8	66.6	63.3	63.0	60.4	56.7	54.2
	2%	DB	52.7	53.2	56.0	61.3	68.4	72.6	74.7	68.5	66.9	61.5	56.9	54.8
		MCWB	51.1	50.6	50.7	53.6	60.0	61.4	63.5	61.7	60.6	58.4	54.8	52.5
	5%	DB	51.2	51.8	53.4	57.9	64.0	68.7	71.0	66.4	64.4	59.7	55.0	53.1
		MCWB	49.6	49.7	49.2	51.6	56.2	60.0	62.3	60.1	59.4	56.9	52.9	51.1
	10%	DB	49.6	50.2	51.4	55.2	60.6	65.7	67.6	64.6	62.2	58.2	53.1	51.6
		MCWB	47.9	48.2	48.1	50.1	54.3	58.4	60.5	59.2	57.9	55.4	51.0	50.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.5	52.2	53.0	57.7	63.4	64.7	67.9	65.1	64.6	61.2	57.3	54.8
		MCDB	54.9	53.9	56.7	64.6	71.7	75.1	77.3	69.6	68.8	63.1	58.5	56.5
	2%	WB	51.5	51.2	51.3	54.6	60.6	62.7	65.4	63.4	62.1	59.0	54.9	53.0
		MCDB	52.5	52.9	54.7	59.7	67.9	71.2	72.5	67.1	65.2	60.8	56.3	54.5
	5%	WB	50.0	50.1	50.0	52.6	57.7	60.9	63.5	61.8	60.5	57.4	53.2	51.6
		MCDB	51.1	51.9	53.2	57.0	62.7	67.2	68.9	65.3	63.3	59.6	54.8	53.0
	10%	WB	48.3	48.5	48.4	50.9	55.5	59.2	61.6	60.1	58.9	55.8	51.3	50.1
		MCDB	49.7	50.3	51.4	54.9	59.2	64.5	66.3	63.8	61.7	57.8	53.0	51.8
Mean Daily Temperature Range	5% DB	MDBR	9.6	10.7	13.6	14.8	14.7	14.9	13.8	12.9	12.7	11.8	10.5	9.8
		MCDBR	10.3	10.0	15.4	18.0	20.2	21.8	19.5	14.3	14.7	11.0	10.0	10.2
		MCWBR	9.9	8.9	11.2	12.0	11.9	12.6	10.4	8.8	10.1	8.8	9.2	9.6
	5% WB	MCDBR	10.5	9.9	14.4	16.5	17.9	18.6	16.6	12.8	12.2	10.4	9.9	10.0
		MCWBR	10.4	9.0	11.5	11.6	11.1	11.5	10.4	9.1	9.3	8.9	9.5	9.6
Clear-Sky Solar Irradiance	taub	0.339	0.348	0.371	0.379	0.390	0.376	0.380	0.362	0.356	0.345	0.332	0.329	
	taud	2.237	2.292	2.259	2.271	2.277	2.348	2.344	2.407	2.419	2.402	2.294	2.204	
	Ebn at Noon	194	236	256	270	272	278	274	273	260	236	197	172	
	Edh at Noon	21	27	34	38	39	37	37	33	29	24	20	18	
All-Sky Solar Radiation	RadAvg	226	443	793	1230	1533	1608	1452	1230	908	532	279	171	
	RadStd	22	37	79	117	106	147	122	95	59	45	26	22	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page